

MERGE 4 DESIGN LLC × L.E.E.P., INC.

FORWARD HOME BUILDING SYSTEM

Project Specification — Four-Tier Performance Framework

Code-Compliant · Net-Zero · Net-Positive · Off-Grid Harvest | City of Wilmington Forward Home Program

TIER 1 Forward Code+ Exceeds IECC 2021 \$220K–\$250K target	TIER 2 Forward Zero Net-Zero Energy ~\$12K–\$18K premium	TIER 3 Forward Positive Net-Positive + Harvest Roof ~\$28K–\$42K premium	TIER 4 Forward Harvest Off-Grid · Agrivoltaic · Food-Producing ~\$52K–\$72K premium
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Merge 4 Design LLC A Wilmington Delaware Mission Driven Partnership merge4design.com · david4d@merge4design.com	L.E.E.P., Inc. Local Economic Empowerment Program Gwenevere Motley, Executive Director Trained installation workforce · Forward Home labor pipeline
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Submitted: May 2026 · Companion files: M4D_ForwardHome_Proforma_May2026.xlsx · M4D_ForwardHome_ProjectDevelopment_May2026.docx

1. How to Use This Specification

This specification defines four escalating tiers of the Forward Home Building System, jointly delivered by Merge 4 Design LLC (design, predevelopment, cost intelligence) and L.E.E.P., Inc. (trained installation workforce). Every tier shares the same Forward Panel wall system and CLT/light-frame structural platform. Tiers escalate by adding energy, food production, and autonomy systems. The tier is selected at predevelopment before schematic design. All four tiers are designed to be deliverable by the L.E.E.P. contractor workforce pipeline.

Tier 1 — Forward Code+: Base standard. Non-combustible wall panel, cold-climate heat pump, fiber cement cladding, double-pane tilt-turn windows, R-49 attic, R-19 floor. HERS 55–70. Energy Star automatic. This tier out-performs the national production builder average at the forward housing price point.

Tier 2 — Forward Zero: Net-zero energy. Triple-pane Crystal Magnus 4500 windows, Quilt 3-zone heat pump (SEER2 25.3/HSPF2 12), Vitoair ERV, heat pump water heater, 5.5 kW PV, SPAN MAIN 40 smart panel. HERS 0–30. DOE Zero Energy Ready Home eligible. ~\$12K–\$18K premium.

Tier 3 — Forward Positive: Net-positive with productive roofscape. Superior Walls Xi Plus foundation, Crystal Magnus 4700 Passive House windows, Mitsubishi FX-Series H2i sumo or GSHP, ERV, 8.5 kW PV, Enphase IQ Battery 5P or Tesla Powerwall 3, LiveRoof MAXX with productive edible planting rows (Up Top Acres). HERS negative to +10. PHIUS+ eligible. ~\$28K–\$42K premium.

Tier 4 — Forward Harvest: Off-grid capable with agrivoltaic roofscape and closed food/water/energy loop. Quilt heat pump optimized for battery draw, EcoFlow DELTA Pro Ultra + FranklinWH aPower 2 battery stack, SPAN MAIN 40 island-mode management, Crop Circle Farms® modular rooftop food system, bifacial TOPCon agrivoltaic trellis (8.5–12 kW above growing beds), 500-gal cistern + drip irrigation, NatureMill composter. ~\$52K–\$72K premium. No utility dependency required.

L.E.E.P. LABOR: All four tiers are designed for installation by the L.E.E.P. contractor cohort trained through the M4Di Academy Forward Home workforce program. Panel installation, envelope sealing, HVAC commissioning, and rooftop system installation are all included in the L.E.E.P. scope. L.E.E.P. field crews are the primary competitive advantage of this system — trained, local, mission-aligned.

QUALITY CONTEXT: The national average HERS score for new US production homes in 2024 was 55 (RESNET data). The Forward Home Tier 1 delivers HERS 55–70 automatically via SIP construction — matching the national best-in-class average — at a \$220K–\$250K forward housing price point that no production builder achieves anywhere in the US today. Tier 2 beats KB Home’s national average (HERS 45) by 15+ points. This is the quality gap that warrants the upgrade conversation.

2. Full Four-Tier Specification

Specification element	TIER 1 Forward Code+	TIER 2 Forward Zero	TIER 3 Forward Positive	TIER 4 Forward Harvest
01 00 00 — GENERAL REQUIREMENTS				
Code basis	2021 IBC/IRC/IECC City of Wilmington Ch. 48	Same + ASHRAE 90.1-2019 DOE ZERH program	Same + PHIUS+ 2021	Same + off-grid island mode design
Sustainability target	Energy Star auto HERS 55–70	DOE Zero Energy Ready Home HERS 0–30	PHIUS+ or PHIUS Zero HERS negative to +10	Off-grid capable Net energy positive Food + energy producing
Blower door target	< 2.0 ACH50 (auto)	< 1.0 ACH50	< 0.6 ACH50 (PHIUS)	< 0.6 ACH50 — critical for battery efficiency
L.E.E.P. scope	Panel install, envelope, HVAC	Above + blower door testing + ERV commissioning	Above + PHIUS rater coordination	Above + agrivoltaic trellis + cistern + Crop Circle install
03 30 00 — FOUNDATION				
Slab / foundation	3,500 psi + SCMs 15-mil vapor retarder Radon stub-outs R-10 EPS under + R-10 edge	Superior Walls® Xi Plus TER 2211-02 (DrJ/ANAB) Built-in EPS+polyiso Mass wall credit IECC R402.2.6 No additional dampproofing	Same as Tier 2 Geothermal well field coordination at footing phase	Same as Tier 3 Well field or enhanced ASHP Cistern footing integrated in slab design
Superior Walls spec	Standard CMU or poured R-10 under + R-10 edge	Xi Plus: 4½" EPS + ½" polyiso 5,000 psi @ 24" OC studs 7,500 lb/ft compression 2-hr ASTM E119 (2× ⅝" Type X) Certified installer required	Same	Same

06 17 13 — STRUCTURAL SPINE + PARTY WALL				
Structural spine	Type A/B: 2×6 SPF-S @ 16" OC Type C/D: TerraLam TL500S 5-ply CLT ESR-5053 (ICC-ES)	Same	Same + embodied carbon EPDs	Same
Party wall	8" CMU continuous (foundation to underside of SIP roof) AHJ-accepted Wilmington	8" CMU + Thermafiber 4" + Hilti FS-ONE MAX both faces 1-hr assembly confirmed	8" CMU + Huber EXACOR® MGO face ESR-4635 2-hr ASTM E119	Same as Tier 3
Acoustic	STC ≥ 50 (CMU inherent)	STC ≥ 55 + acoustic mat	STC/IIC ≥ 60; third-party verified	Same
06 17 53 — 4WARD-SIB FORWARD PANEL (all tiers)				
Wall panel face	Barricade® Thermo-Brace® Blue SIB™ R-5 GPS (Neopor) foam TER 1507-09 (DrJ/ANAB) Structural: 24" OC, 200 mph Vult WRB both faces GREENGUARD Gold	Same — foam inward for CI continuity	Same + interior air barrier membrane (Pro Clima Intello+ or equal)	Same
Cavity insulation	2×6 SPF-S @ 16" OC R-23 blown fiberglass	R-23 Rockwool Comfortbatt (mineral wool — fire + vapor)	Same	Same
Whole-wall R (effective)	R-21 to R-23	R-22 to R-24	R-24 to R-26	R-24 to R-26
Fire-rated party wall face	4Ward-Fire Panel: Huber EXACOR® MGO ESR-4635; 2-hr ASTM E119 Non-combustible ASTM E136	Same	Same	Same
07 27 00 — AIR BARRIERS + WRBs				
Primary WRB	Thermo-Brace SIB — integrated No separate house wrap IRC R703.2 / IBC 1403.2	Same + fluid-applied at all penetrations Prosoco R-Guard or equal	Same + continuous air barrier membrane lapped at foundation and roof	Same — critical for battery/grid autonomy
ACH50 target	< 2.0 (auto SIP)	< 1.0 (verified mid + final)	< 0.6 (PHIUS)	< 0.6
07 46 — CLADDING (Allura® Fiber Cement — all tiers)				
Primary cladding	Allura® Lap Siding 7" exposure; factory primed 50-yr limited warranty Fire/moisture resistant Allura	Allura® Panel — smooth or cedar Allura Soffit all overhangs Stainless rainscreen clips Allura Spectrum	Same + Allura® Shakes accent zones All Spectrum pre-finished	Same as Tier 3

	Trim at all openings 1" ventilated rainscreen on SIB	pre-finished (no field paint)		
08 50 00 — FENESTRATION				
Window system	Crystal Windows Magnus™ 4500 uPVC tilt-turn + hopper 6-chamber frame Double-pane LowE ³ + argon U ≤ 0.20; STC up to 41 dB PG70; HVHZ approved 45% more efficient than thermally broken aluminum	Magnus™ 4500 — triple-pane LowE ³ + krypton U ≤ 0.18 net-zero target SHGC south ≥ 0.30 / east-west ≤ 0.25	Crystal Windows Magnus™ 4700 Passive House certified (REHAU RAU-FIPRO) Triple-pane; U ≤ 0.14 SHGC per PHIUS solar gain limits	Same as Tier 3
Door system	Magnus 4500 swing/tilt-slide U ≤ 0.28	Magnus 4500 tilt-slide Triple-pane; U ≤ 0.20	Magnus 4700 PH certified U ≤ 0.16	Same
23 00 00 — HVAC + VENTILATION				
Space heating/cooling	Standard cold-climate mini-split SEER2 ≥ 18 / HSPF2 ≥ 10 Ductless Heating range to -5°F	Quilt 3-Zone Heat Pump SEER2 25.3 / HSPF2 12 100% capacity @ -5°F 88% @ -13°F 52 dBA outdoor 7½" indoor unit (above doors/windows) OTA firmware upgrades Occupancy sensing room-by-room Energy Star Cold Climate / Most Efficient 2026	Mitsubishi FX-Series H2i sumo™ SEER2 29.9 (single-zone) HSPF2 ≈ 12 100% capacity @ 5°F; -13°F min 46 dBA outdoor R454B; 10-yr warranty — or — GSHP Viessmann Vitocal 222-S COP ≥ 4.5; geothermal loop where site favorable	Quilt 3-zone optimized for battery draw Occupancy sensing reduces island-mode draw OTA load-shedding integration with SPAN MAIN 40
Ventilation	ASHRAE 62.2 min Spot exhaust only	Viessmann Vitoair FS-ERV 85% SRE; MERV 13 CO ₂ -responsive boost ASHRAE 62.2 + 25%	Same + demand-controlled PM2.5 monitoring PHIUS rate compliant	Same
DHW	HPWH; UEF ≥ 3.5 Drainwater heat recovery	Same + solar thermal preheat stub-out	Vitocal 222-S DHW integration Solar thermal preheat Drainwater heat recovery required	Same + cistern preheat loop
26 32 13 — SMART ENERGY + PV + STORAGE				
Smart panel	200A future-ready (EV + PV stub-outs)	SPAN Panel MAIN 40 First UL 3141 certified smart panel (2025) 32 individually controllable circuits Battery-aware outage prioritization Enphase + Tesla	Same	SPAN MAIN 40 — island-mode configured Load-shedding priority schedule: 1. HVAC 2. Fridge 3. Circuits Off-grid ready firmware

		+ SolarEdge compatible Eaton-backed (Q2 2026) \$200–\$600/yr optimization savings		
PV system	Not included (stub only)	5.0–6.5 kW per unit Mono PERC or TOPCon Enphase IQ8 microinverters Utility interconnect + net metering	8.5 kW per unit High-efficiency TOPCon or HJT (≥ 22%) Enphase IQ8 microinverters	8.5–12 kW agrivoltaic trellis Bifacial TOPCon elevated 5–6 ft E/W orientation over growing beds Crops underneath: lettuce, herbs, microgreens
Battery storage	Not included	Enphase IQ Battery 5P (stackable) 15-yr warranty; NEMA 3R — or — Tesla Powerwall 3 (13.5 kWh) Built-in inverter; island-capable	Enphase IQ Battery 5P stack (13–20 kWh) — or — Tesla Powerwall 3 — island capable	EcoFlow DELTA Pro Ultra + FranklinWH aPower 2 19.2–27 kWh base; expandable to 90 kWh 7.2 kW continuous; generator compatible True off-grid island mode
32 93 43 — ROOFTOP / VEGETATED ROOF				
Roof system	Standard EPDM or TPO Ventilated attic R-49 blown-in (CLT deck: R-34–38 above-deck CI + UA alternate path)	LiveRoof® Standard 4.25" soil depth; ±20 lbs/ft² dry FM 4477; ASTM E108 Class A Wind uplift 200 PSF (133 PSF design) Sedum mats Stormwater: 1.75" / 1.1 gal/ft²	LiveRoof® MAXX 8" soil depth; ±40 lbs/ft² dry RoofBlue™ riser: 3.9 gal/ft² storm mgmt Pollinator zones + fire breaks Edible planting rows (Up Top Acres) — lettuce, herbs, cherry tomatoes, kale Subsurface drip; Rain Bird XFS-CV EOR structural confirm required	Agrivoltaic productive roofscape: Crop Circle Farms® modular system (patented; 40+ country trademark) Bifacial PV trellis above growing beds 500-gal cistern; gravity-fed drip NatureMill composter (indoor, 5W) 3-season yield: 200–400 lbs/unit/yr Fire breaks: sedum retained between crop zones
Stormwater	Municipal drains + civil overflow	LiveRoof Standard 1.75" event	LiveRoof MAXX 3.15" + cistern stub	500-gal cistern closed loop Rainwater → crops → compost → soil amendment
CERTIFICATIONS + DOCUMENTATION				
Certifications	Energy Star (auto — SIP) HERS 55–70	DOE Zero Energy Ready Home HERS 0–30; LEED pathway	PHIUS+ 2021 or PHIUS Zero DOE ZERH; LEED Platinum pathway	Off-grid performance verification Food production documentation Net energy positive metered audit

IRA / tax credits	Energy Star appliances	\$45L: \$5,000/unit (DOE ZERH) \$25C HP credits for buyers 30% ITC on PV (developer)	Same as T2	Same + battery ITC (30%)
M4Di documentation	G0 ROM + ALDi Tracker OED cost-verification standard	Same + HERS coordination DOE program enrollment	Same + PHIUS rater Full commissioning plan 01 81 13	Same + annual energy/food production report

3. Premium Cost Summary — Per Unit

All figures G0 ROM ±25% for a 1,300 SF unit on a Type D 10-unit lot cluster. Premiums are incremental above the Tier 1 base. L.E.E.P. labor included throughout.

Cost element	T1 Code+	T2 Zero	T3 Positive	T4 Harvest
4Ward-SIB Forward Panel system (all tiers)	\$13,650	\$13,650	\$13,650	\$13,650
Foundation upgrade — Superior Walls Xi Plus	Base CMU	+ \$4,200–\$6,500	+ \$4,200–\$6,500	+ \$4,200–\$6,500
Cavity insulation upgrade — mineral wool	Std batt	+ \$800–\$1,200	+ \$800–\$1,200	+ \$800–\$1,200
Interior air barrier membrane (Intello+)	—	—	+ \$1,400–\$2,200	+ \$1,400–\$2,200
Cladding — Allura Lap (base all tiers)	\$8,450	\$8,450	\$8,450	\$8,450
Cladding upgrade — Allura Panel + Shakes	Base	+ \$1,200–\$1,800	+ \$2,400–\$3,600	+ \$2,400–\$3,600
Windows — Crystal Magnus 4500 double-pane	\$7,800–\$9,200	Base (included)	Upgrade to 4700	Upgrade to 4700
Window upgrade — triple-pane (4500) or 4700	—	+ \$2,400–\$3,600	+ \$4,800–\$6,400	+ \$4,800–\$6,400
HVAC — Quilt 3-zone vs. standard	Standard mini-split	+ \$2,800–\$4,200	—	Quilt (battery-optimized)
HVAC — Mitsubishi FX-Series H2i sumo	—	—	+ \$4,200–\$6,000	—
HVAC — GSHP (where site favorable)	—	—	+ \$8,000–\$14,000 (alt)	—
Viessmann Vitoair ERV	—	+ \$2,200–\$3,200	+ \$2,200–\$3,200	+ \$2,200–\$3,200
Heat pump water heater (all tiers base)	\$1,800–\$2,400	\$1,800–\$2,400	\$1,800–\$2,400	\$1,800–\$2,400
Drainwater heat recovery	—	+ \$600–\$900	+ \$600–\$900	+ \$600–\$900
SPAN MAIN 40 smart panel	—	+ \$3,500–\$4,500	+ \$3,500–\$4,500	+ \$3,500–\$4,500 (island config)
PV system (5.5–12 kW)	— (stub only)	+ \$15,000–\$19,500	+ \$22,500–\$29,000	+ \$22,500–\$34,000 (agrivoltaic)
Battery storage	—	+ \$9,500–\$13,500 (Enphase/Powerwall)	+ \$9,500–\$13,500	+ \$14,000–\$22,000 (EcoFlow + FranklinWH)
LiveRoof / rooftop system	Std membrane	+ \$5,600–\$9,100 (Standard 700 SF)	+ \$10,500–\$17,500 (MAXX + edible rows)	+ \$14,000–\$22,000 (Agrivoltaic + Crop Circle)

Cistern + drip irrigation	—	—	+ \$1,200–\$2,000	+ \$3,000–\$5,000
NatureMill composter (per unit)	—	—	—	+ \$800–\$1,200
Commissioning / HERS / PHIUS / rater	M4Di ALDi only	+ \$1,200–\$1,800 (HERS)	+ \$3,500–\$5,000 (PHIUS)	+ \$2,500–\$4,000 (off-grid verify)
TOTAL INCREMENTAL PREMIUM (above T1)	— (T1 base)	~\$12,000–\$18,000	~\$28,000–\$42,000	~\$52,000–\$72,000
Estimated sale price range	\$220K–\$250K	\$232K–\$268K	\$248K–\$292K	\$272K–\$322K
IRA \$45L credit (developer)	—	–\$5,000/unit	–\$5,000/unit	–\$5,000/unit
Net to buyer after \$45L	\$220K–\$250K	\$227K–\$263K	\$243K–\$287K	\$267K–\$317K

NEIGHBORHOOD VALUE LIFT: These prices are positioned at the low tier of the current market. Forward Home quality — non-combustible panels, tilt-turn windows, fiber cement cladding, 30-year warranted assemblies — immediately establishes a new comparable in any neighborhood where these homes are built. Within 12–24 months of first occupancy, assessed values in the surrounding blocks will reflect the new standard. The upgrade tiers are defensible on Day 1 and increasingly justified as the neighborhood comp base rises.

4. Performance Summary by Tier

Performance attribute	T1 Code+	T2 Zero	T3 Positive	T4 Harvest
HERS index	55–70	0–30	Negative to +10	Negative (net positive)
ACH50 blower door	< 2.0 (auto)	< 1.0	< 0.6 (PHIUS)	< 0.6 (island efficiency)
Whole-wall R effective	R-21 to R-23	R-22 to R-24	R-24 to R-26	R-24 to R-26
Attic / roof R	R-49 blown-in	R-49–60	R-49+ / LiveRoof MAXX	Agrivoltaic trellis above crops
Window U-factor	≤ 0.20	≤ 0.18 triple-pane	≤ 0.14 (4700 PH)	≤ 0.14
HVAC SEER2 / HSPF2	≥ 18 / ≥ 10	25.3 / 12 (Quilt)	29.9 / ~12 (Mitsubishi)	25.3 / 12 (Quilt battery-optimized)
Annual utility cost	\$900–\$1,400	\$200–\$500	~\$0 to credit	Off-grid — utility bill: \$0
Annual energy savings vs. stick	\$1,800–\$3,200	\$2,800–\$4,500	\$4,500–\$7,000+	\$5,000–\$8,000+ (no utility bill)
Food production	—	—	200–400 lbs/unit/yr (T3 edible rows)	200–600 lbs/unit/yr (agrivoltaic full system)
Stormwater managed	Roof drain	LiveRoof 1.75" event	LiveRoof 3.15" + cistern stub	Closed loop: cistern → crops → compost
Grid dependency	Grid-connected	Grid-connected	Grid-connected (PV export)	Off-grid capable (no dependency)
Certification	Energy Star auto	DOE Zero Energy Ready Home	PHIUS+ / PHIUS Zero	Off-grid performance verified
IRA \$45L (developer)	—	\$5,000/unit	\$5,000/unit	\$5,000/unit
Sale price (net after \$45L)	\$220K–\$250K	\$227K–\$263K	\$243K–\$287K	\$267K–\$317K
L.E.E.P. labor scope	Panel + HVAC	Above + blower door + ERV	Above + PHIUS coordination	Above + agrivoltaic + cistern + Crop Circle

5. Specified Product Directory

5A. Wall Panel — Barricade® Thermo-Brace® SIB™ (all tiers)

Manufacturer: Barricade Building Products / INDEVCO North America, 10351 Verdon Rd, Doswell VA 23047 | barricadebp.com

Evaluation: TER 1507-09 (DrJ Engineering / ANAB ISO/IEC 17065 #1131). Revised May 2023 — verify renewal status at drjcertification.org before permit submission.

- **Product:** Thermo-Brace® Blue SIB™, R-5. 0.120" cellulosic fiberboard + BASF Neopor® GPS foam, reversible orientation.
- **Structural:** WSP equivalent; 370–580 plf ASD wind; R=6.5 SDC A–F; 400 plf uplift; 200 mph Vult.
- **WRB/AIB:** IRC R703.2 / IBC 1403.2; both faces integrated.
- **Energy:** IECC 2021 CZ4 R-5ci satisfied. GREENGUARD Gold.

5B. Foundation — Superior Walls® Xi Plus (Tiers 2–4)

Manufacturer: Superior Walls of America, 937 E Earl Rd, New Holland PA 17557 | superiorwalls.com

Evaluation: TER 2211-02 (DrJ/CBI/ANAB). Revised April 13, 2026; subject to renewal October 1, 2026.

- **Product:** Xi Plus — 1¾" face shell + 4½" EPS + ½" polyiso. 5,000 psi concrete studs @ 24" OC.
- **Structural:** 7,500 lb/ft compression. 235 psf transverse. 745 lb/ft racking (ASTM E72). No dampproofing required.
- **Fire:** 2-hr ASTM E119 with 2× ⅝" Type X GWB. Thermal barrier exemption per IBC 2603.9.
- **Mass wall:** IRC N1102.2.6 / IECC R402.2.6 classified. Heat capacity 6.46 Btu/ft²-°F.
- **Seismic limit:** SDC A/B (IBC); SDC A/B/C (IRC). EOR confirms site seismic assignment.

5C. Cladding — Allura® Fiber Cement (all tiers)

Manufacturer: Allura USA | allurausa.com

- **Allura Lap:** 7"–8" exposure; factory primed; 50-yr warranty. Tier 1 standard.
- **Allura Panel:** Smooth or cedar texture 4×8/4×10; primary field cladding Tiers 2–4.
- **Allura Shakes:** Staggered-edge accent zones at entry/gable; Tiers 3–4.
- **Allura Spectrum:** Pre-finished; no field painting. Complete system: Lap/Panel/Trim/Soffit/Backer.
- **Fire/code:** ASTM E84 Class A; non-combustible cladding layer. 1" ventilated rainscreen on SIB face required.

5D. Fenestration — Crystal Windows Magnus™ Series (all tiers)

Manufacturer: Crystal Window & Door Systems | crystalwindows.com | (800) 472-9988

Magnus 4500 (Tiers 1 & 2): uPVC fusion-welded tilt-turn + hopper. 6-chamber frame, REHAU profile. Dual compression seals, multi-point locks. $U \leq 0.20$ (double) / $U \leq 0.18$ (triple). PG70. STC up to 41 dB. HVHZ approved. 45% more efficient than thermally broken aluminum. LEED/Passive House eligible.

Magnus 4700 (Tiers 3 & 4): REHAU RAU-FIPRO fiber-reinforced profile. Passive House certified. $U \leq 0.14$. Triple-pane. Swing and tilt-slide doors $U \leq 0.16$.

5E. HVAC — Quilt 3-Zone (Tiers 2 & 4); Mitsubishi FX H2i sumo (Tier 3)

Quilt 3-Zone (Tiers 2 & 4): quilt.com. SEER2 25.3 / HSPF2 12. Energy Star Cold Climate / Most Efficient 2026. 100% capacity @ -5°F; 88% @ -13°F. 52 dBA. 7⅞" indoor unit. OTA firmware (23% capacity boost shipped OTA to all installed units). Occupancy sensing. Room-by-room control. SPAN MAIN 40 island-mode integration.

Mitsubishi FX-Series H2i sumo (Tier 3): mitsubishicomfort.com. SEER2 29.9 (12k single). HSPF2 ~12. 100% @ 5°F; -13°F min. 46 dBA outdoor. R454B. 10-yr warranty. Diamond Contractor required.

Viessmann Vitoair FS-ERV (Tiers 2–4): 85% SRE; MERV 13; CO₂-responsive boost; ASHRAE 62.2.

Geothermal (Tier 3 alternate): Viessmann Vitocal 222-S. COP ≥ 4.5 . R32. 150 ft/ton vertical closed-loop. EOR geotechnical confirm required.

5F. Smart Energy — SPAN Panel MAIN 40 (Tiers 2–4)

Manufacturer: SPAN | span.io. First UL 3141 certified smart panel (2025). 32 individually controllable circuits. Battery-aware outage prioritization. Compatible: Enphase + Tesla Powerwall + SolarEdge. EV scheduling. Eaton \$75M investment / joint solutions Q2 2026. Island-mode configuration for Tier 4. \$200–\$600/yr optimization savings.

5G. Battery Storage

Enphase IQ Battery 5P (Tiers 2–3): 5 kWh/module; stackable to 20+ kWh; 15-yr warranty; NEMA 3R; native IQ microinverter integration.

Tesla Powerwall 3 (Tier 3 alt): 13.5 kWh; 11.5 kW peak; built-in inverter; island-capable; gateway load management.

EcoFlow DELTA Pro Ultra + FranklinWH aPower 2 (Tier 4): 19.2–27 kWh base; expandable to 90+ kWh; 7.2 kW continuous; generator compatible; true off-grid island mode. No utility interconnect required.

5H. Rooftop Systems — LiveRoof® + Crop Circle Farms® + Up Top Acres

LiveRoof® Standard (Tier 2): liveroof.com. 4.25" depth; ± 20 lbs/ft² dry; FM 4477; ASTM E108 Class A fire; 200 PSF wind uplift. 1.75" storm retention.

LiveRoof® MAXX (Tier 3): 8" depth; ± 40 lbs/ft² dry; RoofBlue™ riser 3.9 gal/ft² storm mgmt. Edible planting rows integrated by Up Top Acres (uptopacres.com) — lettuce, herbs, tomatoes, kale, strawberries. Subsurface drip: Rain Bird XFS-CV. EOR structural confirmation required.

Crop Circle Farms® (Tier 4): cropcirclefarms.com. Patented modular food-propagation system; 40+ country trademark. Wind-safe anchoring; lightweight planters designed for roof load/drainage. 200+ edible varieties. 3-season yield 200–600 lbs/unit/yr.

Agrivoltaic trellis (Tier 4): Bifacial TOPCon panels elevated 5–6 ft on galvanized trellis above Crop Circle beds. E/W orientation minimizes seasonal shading variability. Shade-tolerant crops below (lettuce, watercress, microgreens) benefit from partial shade in summer heat. Plant transpiration cools underside of panels — mutual performance benefit.

NatureMill composter (Tier 4): Indoor electric; processes food scraps to amendment in 2 weeks. Feeds rooftop growing medium. 120V, 5W average draw. Odor-controlled. Closes the food waste loop within each unit.

6. L.E.E.P., Inc. — Installation Labor Scope

L.E.E.P., Inc. (Local Economic Empowerment Program), Gwenevere Motley, Executive Director, provides trained installation labor for all Forward Home tiers through the M4Di Academy Forward Home workforce program. L.E.E.P. crews represent a core competitive advantage of the Forward Home system: locally trained, mission-aligned workforce that reduces labor cost variability and builds community economic capacity in the neighborhoods where Forward Homes are delivered.

- **All tiers:** 4Ward-SIB panel erection and fastening (TER 1507-09 installation requirements), Allura cladding installation, Crystal window and door installation, HVAC mechanical rough-in and commissioning.
- **Tiers 2–4:** Blower door testing, ERV commissioning, SPAN MAIN 40 installation, PV and battery system installation coordination.
- **Tiers 3–4:** LiveRoof MAXX installation, Up Top Acres productive planting row integration, subsurface drip irrigation.
- **Tier 4:** Crop Circle Farms® modular system installation, agrivoltaic trellis erection, cistern installation and plumbing, NatureMill composter installation.

L.E.E.P. crews are trained by the M4Di Academy in partnership with Merge 4 Design. Training modules cover panel installation, air barrier continuity, blower door protocol, and rooftop system installation. The L.E.E.P. × M4Di Academy contractor cohort is a documented deliverable of the Merge4Ward LLC partnership and is noted in the City of Wilmington OED partnership letter (May 2026).

7. Required Submittals by Tier

All tiers (Tier 1 minimum)

- TER 1507-09 (Thermo-Brace SIB) — submit with permit application
- ESR-4635 (Huber EXACOR, fire panel) — submit with permit application
- Allura fiber cement ESR — submit at permit
- Crystal Magnus 4500 NFRC report — submit at permit
- M4Di ALDi cost report — submit to OED for cost verification

- Blower door results — mid-construction and CO

Tier 2 additions

- TER 2211-02 (Superior Walls Xi Plus) — submit with permit
- Quilt 3-Zone product data + Manual J — submit at permit
- SPAN MAIN 40 spec sheet — submit at permit
- LiveRoof Standard + structural confirmation — submit at permit
- Enphase or Tesla PV/battery shop drawings — submit at permit
- HERS rating documentation — submit at CO
- DOE Zero Energy Ready Home enrollment — submit at preconstruction

Tier 3 additions

- Crystal Magnus 4700 Passive House certification — submit at permit
- Mitsubishi FX-Series H2i sumo or GSHP engineering — submit at permit
- LiveRoof MAXX + RoofBlue + EOR structural confirmation — submit at permit
- Up Top Acres planting plan — submit at permit
- PHIUS rater engagement letter — submit at permit application
- PHIUS commissioning plan — submit at preconstruction

Tier 4 additions

- Crop Circle Farms® installation plan — submit at permit
- Agrivoltaic trellis structural engineering — submit at permit
- Cistern plumbing plan — submit at permit
- EcoFlow/FranklinWH battery island-mode configuration — submit at permit
- SPAN MAIN 40 island-mode load schedule — submit at permit
- Off-grid performance verification plan — submit at preconstruction

8. Contact

Merge 4 Design LLC

A Wilmington Delaware Mission Driven Partnership
merge4design.com · david4d@merge4design.com

L.E.E.P., Inc.

Local Economic Empowerment Program
Gwenevere Motley, Executive Director
Trained installation workforce · Forward Home labor pipeline

Tier selection, predevelopment cost verification, M4Di OI Tool ROM, and ALDi Tracker services: contact Merge 4 Design LLC at david4d@merge4design.com.

L.E.E.P. workforce deployment, contractor cohort coordination: contact L.E.E.P., Inc. at the address above.